

1 NAME OF THE MEDICINAL PRODUCT

Ganaton® Tablet 50mg

2 QUALITATIVE AND QUANTITATIVE COMPOSITION

Each film coated tablet contains 50 mg Itopride Hydrochloride.

For the full list of excipients, see section 6.1.

3 PHARMACEUTICAL FORM

Itopride hydrochloride is a white to pale yellowish white crystal or crystalline powder. It is odorless and has a bitter taste. It is very soluble in water, freely soluble in methanol or in glacial acetic acid, sparingly soluble in ethanol and practically insoluble in acetic anhydride or in ether. The pH of the solution is 4.0 to 5.0. The melting point is 193 to 198°C.

4 CLINICAL PARTICULARS

4.1 Therapeutic Indications

Treatment of gastrointestinal symptoms caused by gastric dysmotility and delayed gastric emptying, like sensation of bloating, early satiety, postprandial fullness, upper abdominal pain or discomfort, anorexia, heartburn, nausea and vomiting; functional (non-ulcer) dyspepsia or chronic gastritis.

4.2 Posology and Method of Administration

Adults

The recommended dose of itopride hydrochloride for adult patients is 150 mg daily [one tablet (50 mg) taken orally three times a day before meals]. The dose may be reduced according to the patient's age symptoms (see section 4.4).

Duration of Treatment:

In clinical studies, itopride hydrochloride has been administered up to 8 weeks.

4.3 Contraindications

Itopride hydrochloride is contraindicated in patients with known hypersensitivity to itopride hydrochloride or any of the excipients.

Itopride hydrochloride should not be used in patients in whom an increase in gastrointestinal motility could be harmful, e.g. gastrointestinal hemorrhage, mechanical obstruction or perforation.

4.4 Special Warnings and Precautions for Use

General

Itopride hydrochloride enhances the action of acetylcholine and may produce cholinergic side effects. Data on long-term use are not available.

Information on excipients known effect:

Itopride hydrochloride 50 mg film coated tablets contains lactose. Patients with rare hereditary problems of galactose intolerance, total lactase deficiency or glucose-galactose malabsorption should not take this medicine.

Usage in Special Populations

Pediatric Use

Safety of this product in children under the age of 16 has not been established.

Geriatric Use

In general, appropriate caution should be exercised in the administration and monitoring of Itopride hydrochloride in elderly patients reflecting the greater frequency of decreased hepatic, renal function, and of concomitant disease or other drug therapy.

4.5 Interaction with Other Medicinal Products and Other Forms of Interaction

Metabolic interactions are not expected since itopride is primarily metabolized by flavine monooxygenase and not by CYP450.

No changes in protein binding have been seen with coadministration of warfarin, diazepam, diclofenac sodium, ticlopidine hydrochloride, nifedipine, and nicardipine hydrochloride.

Since itopride has gastrokinetic effects it could influence the absorption of concomitantly orally administered drugs. Particular caution should be taken with drugs with a narrow therapeutic index, sustained release or enteric-coated formulations.

Anti-ulcer drugs like cimetidine, ranitidine, teprenone and cetraxate do not affect the prokinetic action of itopride.

Anticholinergic drugs may reduce the action of itopride.

4.6 Fertility, Pregnancy and Lactation

Fertility

No human data on the effect of Itopride on fertility are available; however, animal studies do not indicate harmful effects of Itopride (see section 5.3).

Pregnancy

There are no or limited amount of data (less than 300 pregnancy outcomes) from the use of itopride in pregnant women. Animal studies do not indicate direct or indirect harmful effects with respect to reproductive toxicity (see section 5.3). As a precautionary measure, it is preferable to avoid the use of itopride during pregnancy.

Lactation

Itopride is excreted in breast milk of animals, but there is insufficient information on the excretion of itopride in human milk. A risk to the suckling child cannot be excluded. A decision must be made whether to discontinue breast-feeding or to discontinue/abstain from itopride therapy taking into account the benefit of breast feeding for the child and the benefit of therapy for the woman.

4.7 Effects on Ability to Drive and Use Machines

Although, no studies on the effects on the ability to drive and use machines have been performed, adverse drug reactions such as dizziness may occur (see section 4.8). Under this condition the ability to react may be decreased.

4.8 Undesirable Effects

The following undesirable effects have been experienced with the below indicated frequencies in 1559 itopride-treated patients in 4 placebo-controlled, 5 reference-controlled, and 13 uncontrolled company sponsored interventional clinical trials with a standard daily itopride dose of 150 mg or lower [common ($\geq 1/100$ to $< 1/10$) and uncommon ($\geq 1/1,000$ to $< 1/100$)]. No ADRs in the categories very common ($\geq 1/10$), rare ($\geq 1/10,000$ to $< 1/1,000$), or very rare ($< 1/10,000$) were identified.

MedDRA SOC	Frequency category	
	Common	Uncommon
Gastrointestinal disorders	Diarrhea, Abdominal pain	Nausea, Abdominal discomfort, Salivary hypersecretion
Investigations		Alanine aminotransferase increased, White blood cell count decreased, Aspartate aminotransferase increased
Endocrine disorders		Hyperprolactinemia
Nervous system disorders	Headache	Dizziness
Skin and subcutaneous tissue disorders		Rash

ADR - Post Marketing Experience

The following adverse events have been reported spontaneously during post marketing use. A precise frequency cannot be estimated from the available data.

Blood and lymphatic system disorders

Leukopenia and thrombocytopenia.

Immune system disorders

Hypersensitivity, including Anaphylactic reaction.

Nervous system disorders

Dizziness, headache, and tremor.

Gastrointestinal disorders

Diarrhea, constipation, abdominal pain, salivary hypersecretion, and nausea.

Hepato-biliary disorders

Jaundice.

Skin and subcutaneous tissue disorders

Rash, Erythema and Pruritus.

Reproductive system and breast disorders

Gynecomastia.

Investigations

Aspartate aminotransferase increased, Alanine aminotransferase increased, Gamma-glutamyltransferase increased, Blood alkaline phosphatase increased, Blood bilirubin increased, and Blood prolactin increased.

4.9 Overdose

In case of excessive overdose, the usual measures of gastric lavage and symptomatic therapy should be applied.

5 PHARMACOLOGIC PROPERTIES

5.1 Pharmacodynamics Properties

Itopride hydrochloride is an orally administered gastroprokinetic agent. Itopride hydrochloride tablets contain 50mg of itopride hydrochloride. The tablet has been formulated to provide immediate release. It is chemically designated as N-[4-[2-Dimethylamino)ethoxy]benzyl]-3,4-dimethoxybenzamide hydrochloride. Itopride hydrochloride is a substituted benzamide. It has an empirical formula of C₂₀H₂₆N₂O₄.HCl, molecular weight of 394.89 g/mol.

Itopride has dual mechanism of action: dopamine D₂ receptor antagonism and acetylcholinesterase inhibition. The effect of itopride is an increase in acetylcholine concentration, which promotes gastric motility, increases the lower oesophageal sphincter (LOS) pressure, accelerates gastric emptying and improves gastroduodenal coordination.

Itopride hydrochloride also has antiemetic action through interaction with D₂ receptors located in the chemoreceptor trigger zone. This was demonstrated by dose dependent inhibition of apomorphine-induced vomiting in dogs.

In conscious dogs, itopride hydrochloride activates propulsive gastric motility through dopamine D₂receptor antagonistic actions and dose-dependent inhibition of acetylcholinesterase.

Itopride hydrochloride has been shown to accelerate gastric emptying in humans, dogs and rats. The action of itopride hydrochloride is highly specific for the upper gastrointestinal tract.

Itopride hydrochloride does not affect serum gastrin levels.

For functional dyspepsia (FD), the results of 2 placebo-controlled studies revealed statistically significant benefit or clear tendencies for the measured endpoints (Holtmann, 2003 and Yun, 2001). In Holtmann 2003, the symptom severity Leads dyspepsia questionnaire (LDQ) score of the difference between the placebo group and the pooled results of the itopride dose groups (50 mg, 100 mg and 200 mg tid) revealed superiority of itopride (P<0.05). Although the test for a dose response was not statistically significant (P=0.06) and the difference between placebo and itopride in the 50 mg tid dose revealed P=0.07, the other two doses were statistically significant (P=0.05). After 8 weeks, overall 41% of the patients receiving placebo were symptom-free or had marked improvement of their motility disorder compared to 57%, 59%, and 64% receiving itopride of 50, 100, or 200 mg tid, respectively (P<0.05 for all comparisons). The analysis of the combined endpoint of pain and fullness showed that itopride yielded a greater rate of response than placebo

(73% vs. 63%, $P < 0.05$). In Yun 2001, the baseline comparison of the symptom scores for motility disorder measured after 2 weeks: itopride (50 mg tid) 89.1%; placebo: 62.5% demonstrated that itopride was better than placebo for subject's global satisfactory evaluation and investigator's global evaluation, with statistical significance ($P < 0.05$). It showed that itopride was effective in chronic gastritis with dyspepsia.

Overall, the study results of 19 efficacy studies in the category FD showed a positive effect of itopride and patients reported to benefit from itopride treatment. A controlled study of itopride versus mosapride (Amarapurkar 2004), showed a reduction of FD symptoms with both itopride and mosapride treatment and a higher reduction of the severity of symptoms in the itopride group. Statistically significant higher moderate to complete symptom relief was demonstrated with itopride versus metoclopramide (Kamath 1999). In Sawant 1999, patient symptom relief was better for itopride as compared with domperidone.

The results of a meta-analysis of 9 randomized controlled trials enrolling 2,620 FD cases were published by Huang et al. 2012. Compared with control groups (placebo, domperidone, mosapride), itopride had superior relative risk values for global patient assessment, postprandial fullness and early satiety. Itopride improved the LDQ scores more significantly than placebo. The meta-analysis from Wu et al. 2010 of 18 randomized controlled trials in 2,420 subjects showed that the relief rate of anorexia and early satiety in itopride group were statistically superior compared with domperidone.

Studies in diabetic gastroparesis showed that itopride tends to accelerate gastric emptying of liquids and solids in longstanding diabetes (Stevens, 2008). In non-insulin dependent diabetes mellitus patients a positive effect on diabetic gastroparesis (using acetaminophen and radiopaque markers) was found after 2 weeks of itopride (150 mg/day) treatment (Noritake Masayuki, 1997). Alpha-lipoic acid and itopride combination therapy can further improve symptoms of gastroparesis, increase gastric emptying, and decrease the levels of motilin and gastrin compared with treatment with itopride alone (Ma, 2015a). Combination treatment of itopride showed significant improvement in severity as well as the frequency of all the symptom parameters of the disease (Venkatesh, 2008).

Effectiveness of itopride in gastro-esophageal reflux disease (GERD) can in part be explained due to it inhibiting transient lower esophageal sphincter relaxations. (Scarpellini 2011) as well as significantly reducing the % time pH < 4 . The study results in GERD demonstrated that adding itopride to proton pump inhibitors (lansoprazole or pantoprazole) in the treatment of laryngopharyngeal reflux improves symptoms scores (Chun, 2013; Ezzat, 2011) shortens the need for oral medication, and reduces the rates of recurrence of symptoms after discontinuation of treatment (Ezzat, 2011). Itopride was more effective than domperidone in combination with omeprazole (a proton pump inhibitor) in the treatment of GERD in patients with diabetes mellitus (Fedorchenko, 2013) and also in an uncontrolled study symptoms scores improved in patients with GERD after itopride treatment (Choi, 2003).

5.2 Pharmacokinetic Properties

Absorption

Itopride hydrochloride is rapidly and almost completely absorbed from the gastrointestinal tract. Relative bioavailability is calculated to be 60% due to liver first pass metabolism. There is no effect of food on bioavailability. Peak plasma levels (C_{max} 0.28 g/mL) are reached after 0.5 to 0.75 hours after 50 mg of itopride hydrochloride.

Following multiple oral doses ranging from 50 mg to 200 mg tid, itopride hydrochloride and its metabolites showed linear pharmacokinetics over a treatment period of seven days, with minimal accumulation.

Distribution

Approximately 96% of itopride hydrochloride is bound to plasma proteins. Albumin accounts for most of the binding. Alpha-1-acid-glycoprotein accounts for less than 15% of binding.

In rats, itopride is distributed extensively ($V_d = 6.1$ L/kg) with high concentrations in the kidneys, small intestines, liver, adrenal glands, and stomach. Protein binding in the rat was lower than that seen in humans (78 vs 96%). The transfer into the central nervous system, including brain and spinal cord, was minimal. Itopride distributes into the breast milk of rats.

Metabolism

Itopride undergoes extensive hepatic metabolism in humans. Three (3) metabolites have been identified, of which only one exerts minor activity without pharmacological relevance (approximately 2-3% of that of itopride). The primary metabolite in humans is the N-oxide, generated by oxidation of the tertiary amine N-dimethyl group.

Itopride is metabolized by a flavin-dependent monooxygenase (FMO3). The abundance and efficiency of the human FMO-isozymes can be subject to genetic polymorphisms, which can lead to a rare autosomal recessive condition known as trimethylaminuria (fish odor syndrome).

Pharmacokinetics may differ between trimethylaminuria (TMAU) patients and non-trimethylaminuria (non-TMAU).

In vivo pharmacokinetic studies on CYP-mediated reactions revealed that itopride showed neither inhibitory nor inductor effect on CYP2C19 and CYP2E1. CYP content and uridine diphosphate glucuronosyl transferase activity were not altered with the administration of itopride.

Excretion

Itopride hydrochloride and its metabolites are primarily excreted in the urine. The urinary excretions of itopride and its N-oxide were 3.7% and 75.4%, respectively, in healthy subjects after oral administration of a single therapeutic dose (50mg). The terminal phase half-life of itopride hydrochloride was approximately six (6) hours.

5.3 Pre-Clinical Safety Data

Pre-clinical effects were observed only at dosages in excess of human dose indicating little relevance to human use. Moreover, humans are less susceptible to the hormonal effects observed in the animal models. High dosages (30 mg/kg/day) of itopride hydrochloride produced reversible mammary hyperplasia in rats, secondary to hyperprolactinemia, albeit not in dogs (to 100mg/kg/day) or primates (to 300 mg/kg/day). A 3-month oral toxicity study in dogs, prostatic atrophy was observed at 30 mg/kg/day of itopride, albeit not after 6 months treatment with higher (100mg/kg/day) dosages nor in rats or monkeys at even higher (300 mg/kg/day) dosages.

Long-term animal studies of itopride hydrochloride have not been conducted to evaluate carcinogenic potential.

No clastogenic or mutagenic effects of itopride hydrochloride were observed in a battery of invitro

and in vivo assays.

In a rat fertility study, prolonged estrous cycle was seen in female animals receiving ≥ 30 mg itopride/kg/day, secondary to hyperprolactinemia. In addition, a prolonged precoital interval was also seen at 300 mg/kg/day. However, the plasma concentrations of itopride in pregnant rats treated orally at 300 mg/kg were about 100- fold higher than the itopride concentrations observed in human serum after clinical administration (50 mg). There was no adverse effect on copulation or fertility. No teratogenic effect was observed in preclinical studies in rats and rabbits, at maternal doses of up to 300 mg/kg.

6 PHARMACEUTICAL PARTICULARS

6.1 List of Excipients

Inactive Ingredients

Itopride hydrochloride tablets are available for oral administration in a strength of 50 mg with the following inactive ingredients: lactose, corn starch, carmellose, light anhydrous silicic acid, magnesium stearate, hydroxypropyl methylcellulose 2910 (6 mm²/s), macrogol 6000, titanium oxide and carnauba wax.

6.2 Storage

Store below 30°C.

6.3 Shelf Life

36 months.

6.4 How Supplied

Itopride hydrochloride tablet is white, round film-coated tablets with a breaking score on one side and embossed with the code “HC 803” on the other side.

7 MANUFACTURER

Katsuyama Pharmaceuticals K.K., Katsuyama
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8 PRODUCT REGISTRATION HOLDER & IMPORTER

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Revised November 2025

RDCCDS000288/5